Problem 1

(a)

ACC: The general public and the model developer will benefit from optimizing this metric. ACC measures overall correctness, so a higher ACC means that the general public will live in a safer community, and more organizations will use this model.

PPV: Defendants will benefit from optimizing this metric. Optimizing this metric will reduce the likelihood of high-risk people who do not re-offend.

FPR: Defendants will benefit from optimizing this metric. Optimizing this metric will make it less likely to wrongly classify people as high-risk.

FNR: The victims of crimes and public safety will benefit. Optimizing this metric means catching more actual re-offenders and reducing the public safety risk.

SP: Civil rights advocates and people focusing on group fairness will benefit. Optimizing SP ensures no group is systematically more likely to be labeled high-risk. But in the COMPAS context, optimizing SP could reduce predictive accuracy if base recidivism rates differ by group.

(b)

A: Prophecy's training data comes from past hiring data in tech companies, which historically favored men over women due to the biases of past recruiters. Prophecy will then label women as less promising than men. In this case, qualified applicants who identify as women will be harmed by this model. To deal with this, we can audit the training data to find if there is any bias and make it representative of the qualified applicants before training the model.

B: Prophecy uses word embeddings trained on internet text that give Asian names lower scores because they are often on H1B visas. Applicants with Asian names are harmed in this case, even if they do not need sponsorship. We can make the names invisible to the model and use only numbers to identify them.

C: Prophecy is trained by data when Python is the most commonly used language in data science. However, let's say in 2026, R becomes more popular and every data science undergrad starts to learn R. However, Prophecy still ranks Python-heavy resumes higher, disadvantaging recent graduates with strong R skills. Recent graduates and career-changers whose skills mismatch outdated training data are harmed. We can regularly retrain the model using current data, making sure it is not outdated.

Problem 2

(a) d(A,B) = 0.09-0 = 0.09
d(C,D) = 1-0.91 = 0.09
D(M(A), M(B)) = 0.4-0.1 = 0.3 > 0.09
D(M(C), M(D)) = 0.95=0.8 = 0.15 > 0.09
The Lipschitz condition does not hold.

(b) d(A,C) = 0.91-0 = 0.91
d(E,F) = 0.64-0.36 = 0.28
Since the penalty is 0, whether they are in different groups does not matter.
D(M(A), M(C)) = 0.8-0.1 = 0.7 < 0.91

D(M(E), M(F)) = 0.65-0.55 = 0.1 < 0.28
The Lipschitz condition holds.

(c) d(A,C) = min(1, 0.91+0.2=1.11) = 1
d(E,F) = min(1, 0.28+0.2=0.48) = 0.48
D(M(A), M(C)) = 0.8-0.1 = 0.7 < 1
D(M(E), M(F)) = 0.65-0.55 = 0.1 < 0.48
Change in penalty δ didn't change the outcome here. The Lipschitz condition still holds.

(d) Case 1: $\Delta_s + \delta < 1$

$D(M(x), M(y)) \leq \Delta_s + \delta \text{ -> } \delta \geq D(M(x), M(y)) - \Delta_s$.

So the minimal δ in this case is $max(0,\ D(M(x), M(y)) - \Delta_s)$.

Case 2: $\Delta_s + \delta > 1$

Then d(x, y) = 1, and since $D(M(x), M(y)) \leq 1$, the condition always holds.

So the final answer is $\delta^* = \max(0,\ |M(x) - M(y)| - |s(x) - s(y)|)$.

Problem 3

(a) Here is the result of my baseline model:

```
=== Baseline Overall ===
recall       0.000000
accuracy     0.889963
precision    0.000000
fnr          1.000000
fpr          0.000000
dtype: float64

=== Baseline by Gender ===
             recall  accuracy  precision  fnr  fpr
gender_Male
0               0.0  0.888736        0.0  1.0  0.0
1               0.0  0.891410        0.0  1.0  0.0
=== Fairness Metrics (Baseline) ===
False Negative Rate Difference: 0.0000
False Positive Rate Difference: 0.0000
Demographic Parity Ratio: nan
Equalized Odds Difference: nan
Selection Rate Difference: 0.0000
```

The baseline random forest model (with default hyperparameters) achieves a seemingly high accuracy of 89%. However, this metric is misleading due to the severe class imbalance in the dataset: The dataset contains only about 10% positive class samples, that is, readmitted patients. So the model adopts a trivial strategy. It predicts all instances as the majority class (0).

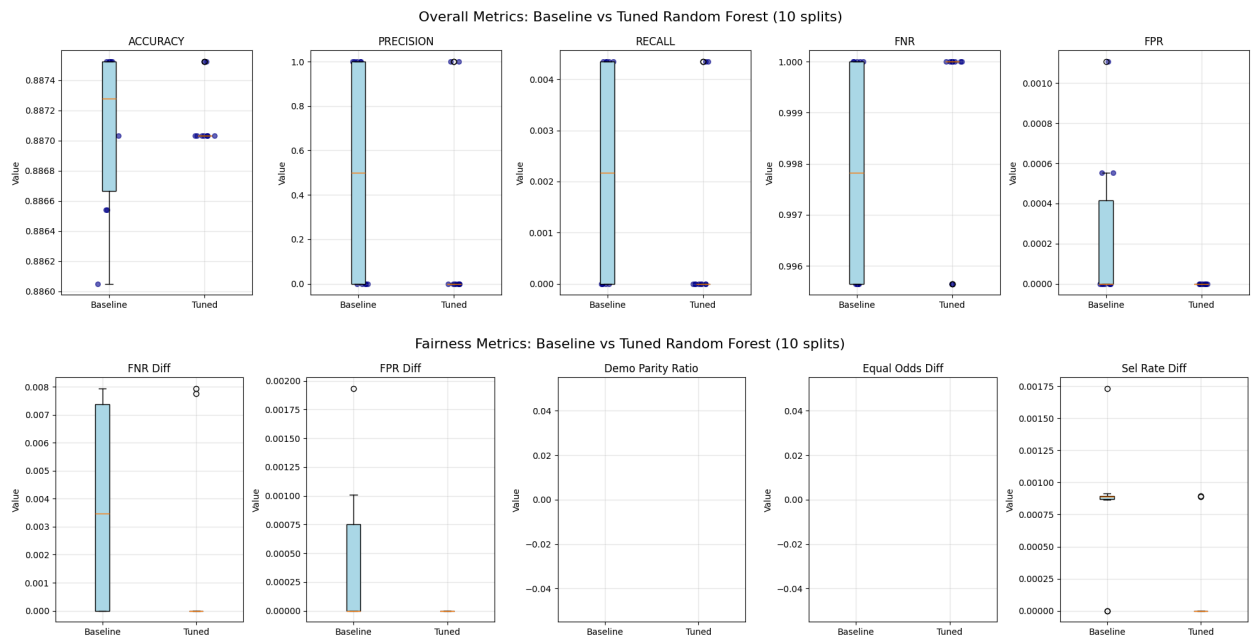
This explains the perfect but undesirable metrics:

- Recall = 0: The model fails to identify any positive cases
- Precision = 0: No positive predictions were made
- FNR = 1.0: All actual positive cases were misclassified as negative
- FPR = 0.0: No false positives (since no positive predictions were made)

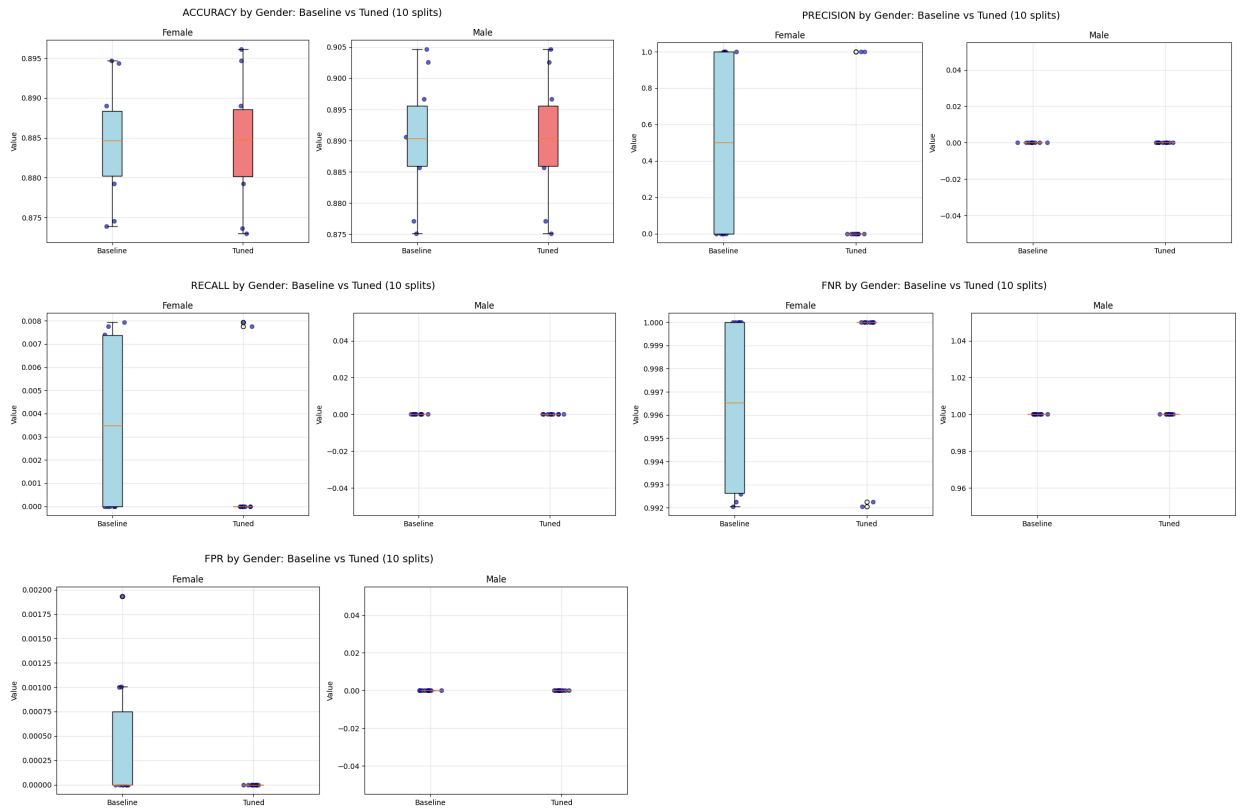
Both gender groups exhibit identical problematic behavior. Both groups have recall = 0 and precision = 0, and FNR is also 1.0 for both groups. Therefore, it is easy to see why FNR difference, FPR difference, and selection rate difference are all 0.

The NaN values for Demographic Parity Ratio and Equalized Odds Difference occur also because there are no positive predictions.

(b) Here's the result of my tuned model:



The baseline model show a high variance in accuracy, but the tuned model is very stable. It's accuracy stays at about 0.887. The variance for the precision and recall in baseline model is also high, but in tuned model it stays at about 0 with some outliers. The extremely low recall suggests the model is failing to identify the positive class, likely due to the severe class imbalance in the dataset. Both models show FNR at about 1, meaning that the model is missing almost all of the positive cases. For the FPR, the baseline model shows slight variation, while the tuned model stays at 0. Compared to the baseline model, the tuned model has low FNR difference, the FPR difference, and the selection rate difference. However, this may still be the result of the model predicting everyone as negative class.

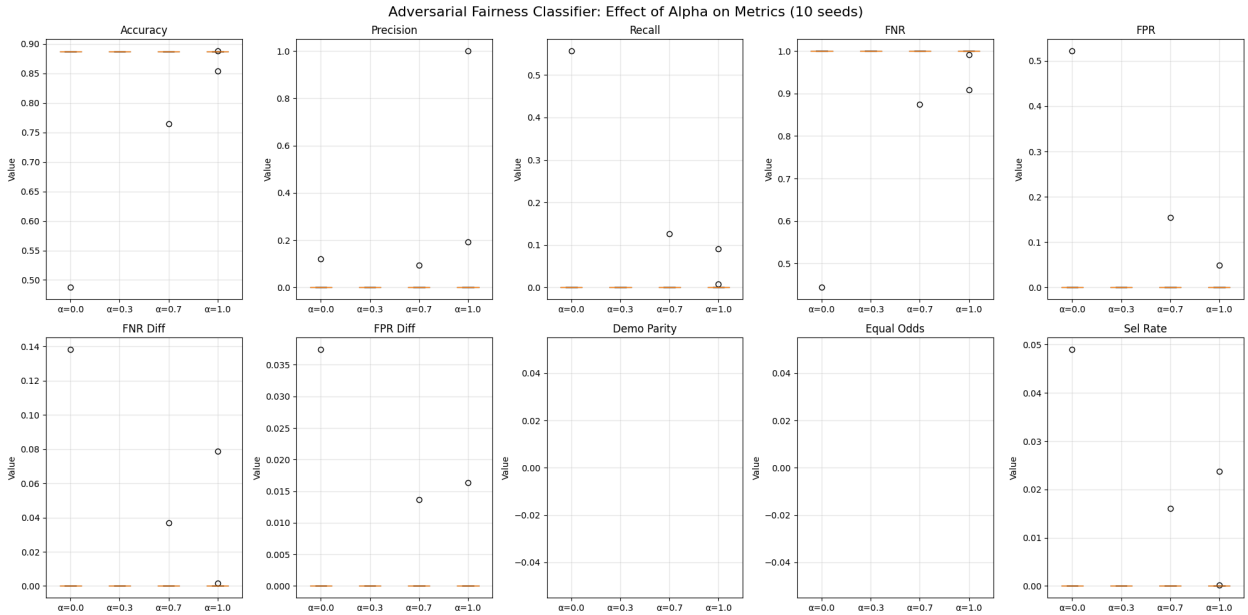


When we look at the group-specific result, we can see that most of the variations of the baseline model come from female sample. The model has a slightly higher accuracy for male than female. Since in some cases the baseline model successfully predicts some true positive female patients, I would suggest the baseline model is performing slightly better for female than male. But I don't think this is very meaningful, since the model is not learning the pattern but trivially predicting almost everyone as the majority class. This variation may also be by chance.

My hypothesis is that the hyperparameter max_depth=10 makes the trees too shallow to learn the true pattern. However, in this highly imbalanced dataset, the model can trivially predict everyone to be the majority group to reach high accuracy. Also, for the n_estimators part, I think when you have 1000 shallow trees, each slightly biased toward the majority class, their collective vote becomes an even stronger bias. This is like asking 1000 people who are all slightly conservative to vote—the outcome is overwhelmingly conservative.

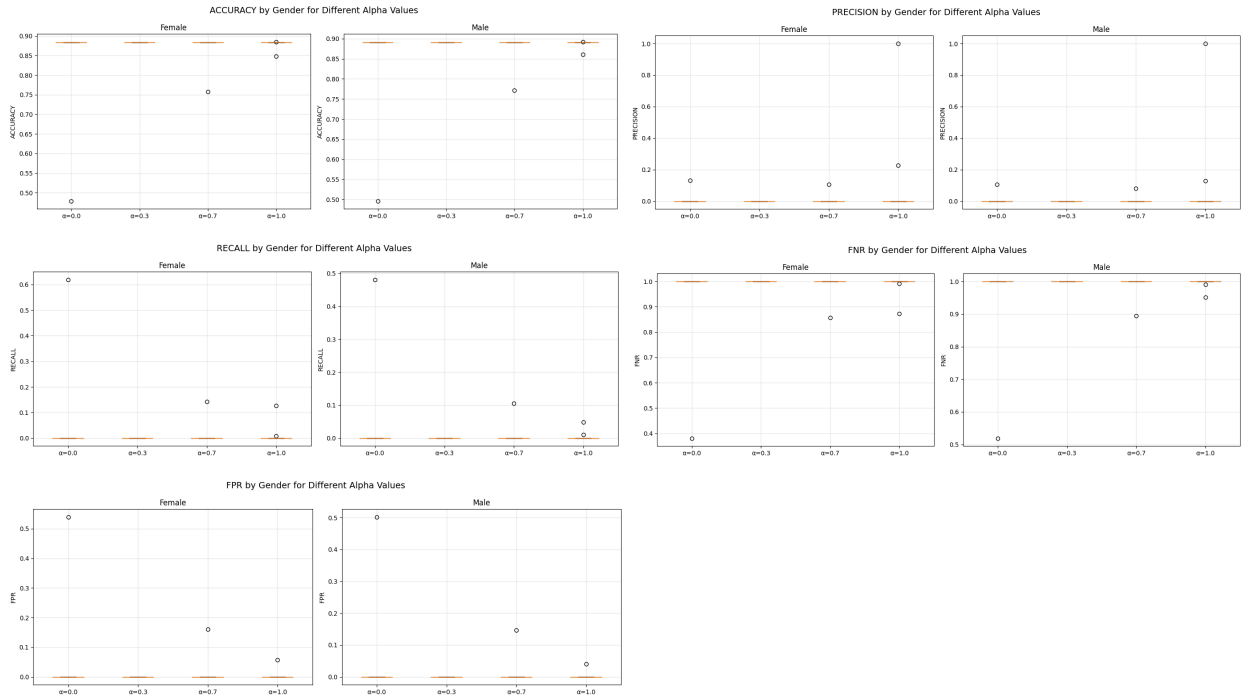
The baseline model showed variation across splits because its default parameters allow it to adapt to the specific characteristics of each training set. With no depth limit, trees can grow to capture patterns for each train-test split, leading to different predictions across splits. The tuned model showed no variation because its constrained architecture (max_depth=10) limits its ability to adapt. When combined with the size (1000 trees), the model converges to the same stable solution regardless of the specific split. It always predicting the majority class.

(c) Here's my result for the Adversarial Fairness Classifier:



Overall, the results show that changing α\alphaα has very limited impact on most metrics. Accuracy remains consistently high (around 0.88–0.89) across all values of alpha, with only one or two outliers due to randomness in training. However, precision and recall are extremely close to zero for most runs, indicating that the classifier rarely predicts the positive class. This behavior is also reflected in the false negative rate (FNR), which is close to 1.0 across almost all values of alpha, suggesting that almost all positive instances are misclassified as negative.

Group-specific values and fairness-related metrics such as demographic parity difference, equalized odds difference, and selection rate difference are also near zero for all values of alpha. While this might indicate strong fairness performance, it is still largely a consequence of the model predicting almost no positive outcomes for any group. When a classifier outputs nearly identical predictions for all individuals, fairness gaps are naturally low, but this does not necessarily imply meaningful fairness improvements.

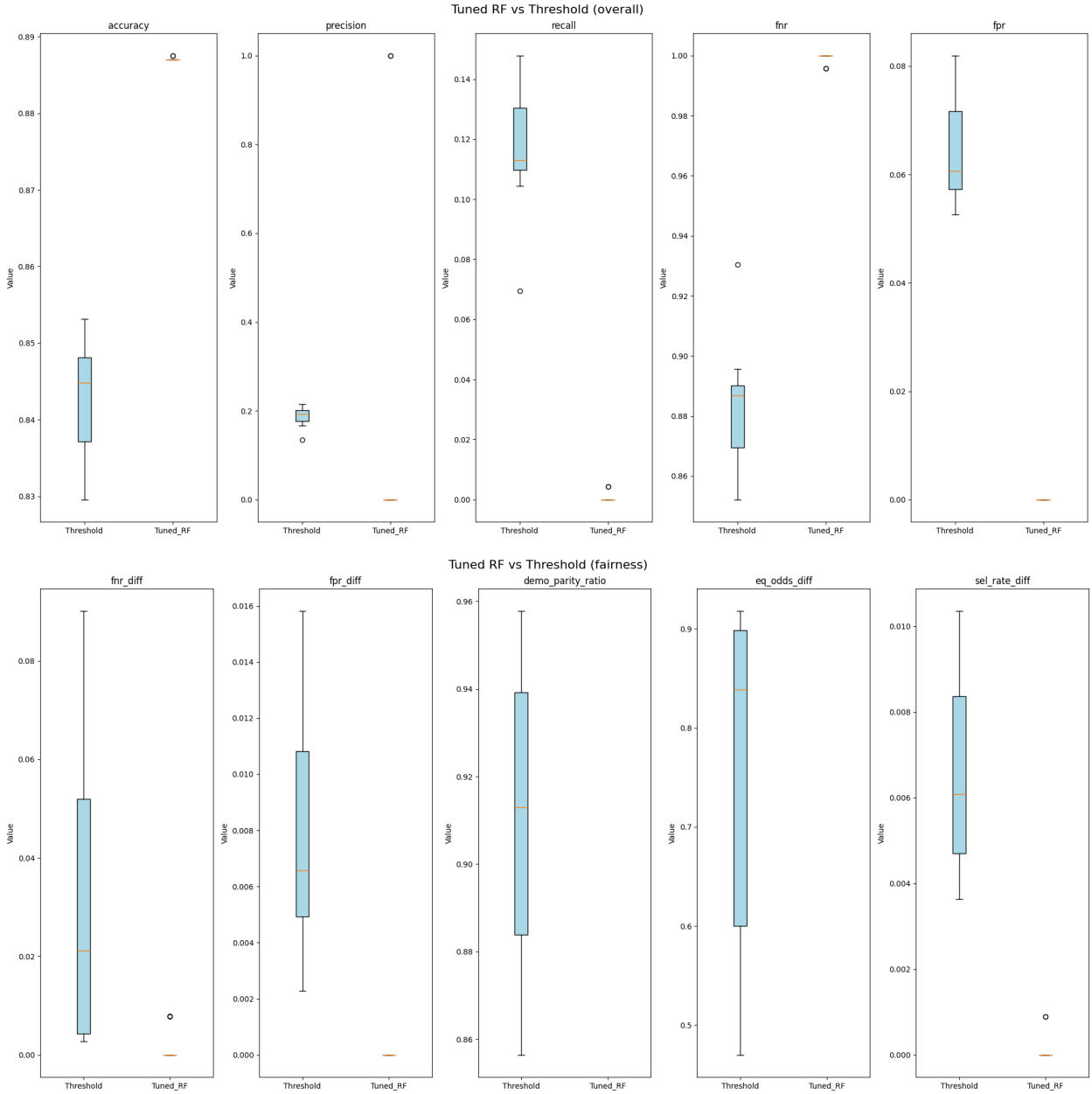


The results from the Adversarial Fairness Classifier in part (c) are broadly consistent with the behavior observed in the tuned Random Forest model from part (b). Both models achieve relatively high accuracy (approximately 0.88–0.89), which is largely due to the strong class imbalance in the dataset. Both models show low recall and high false negative rates. Therefore, I think neither the tuned Random Forest nor the Adversarial Fairness Classifier preforms well on predicting the positive instances. Both models produce very few positive predictions, so outcome distributions across demographic groups naturally become similar, which reduces the disparity but not necessarily means fairness.

Overall, the comparison indicates that the Adversarial Fairness Classifier does not provide clear advantages over the tuned Random Forest for this dataset. Both models are dominated by the class imbalance and converge toward similar prediction patterns characterized by high accuracy but poor recall.

(d)

Here is my result of Threshold Optimizer compared to tuned random forest model in part (b):

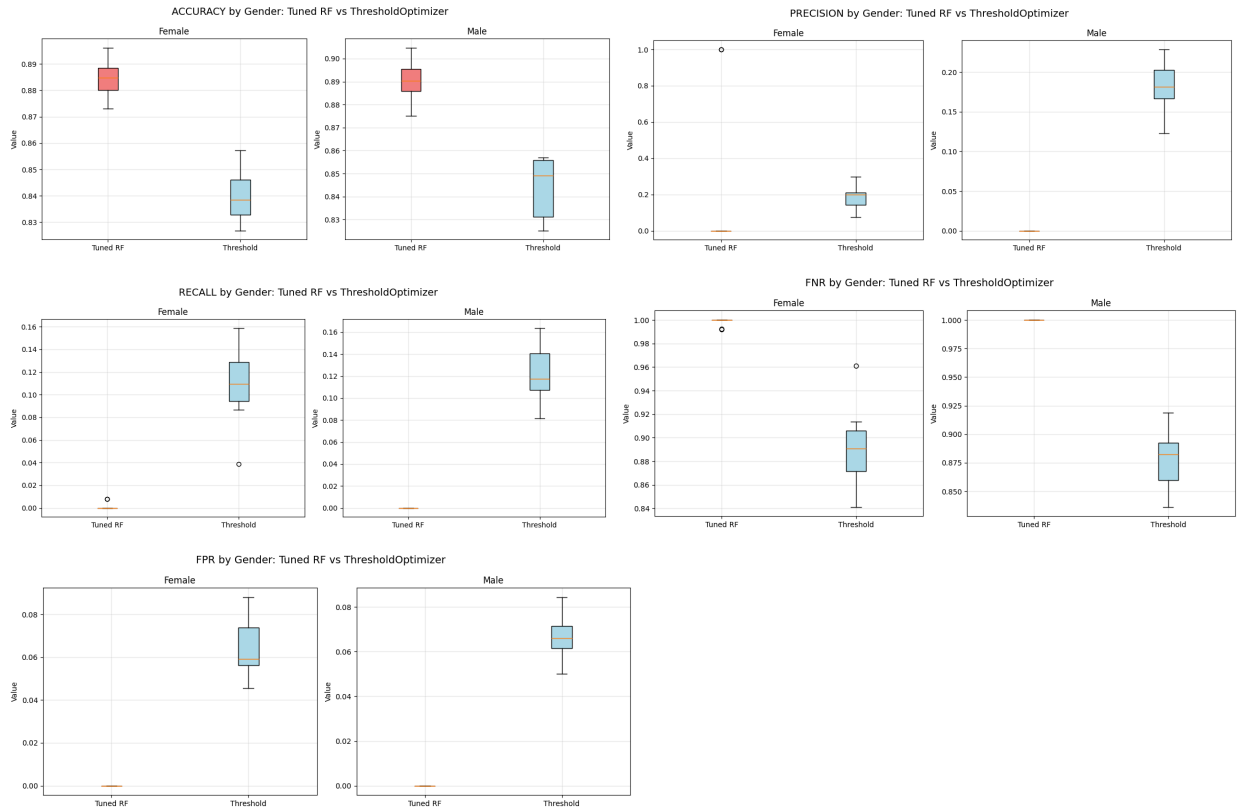


In this medical classification setting, I selected equalized odds as the fairness constraint in the ThresholdOptimizer. Equalized odds requires that both the false positive rate (FPR) and false negative rate (FNR) be similar across demographic groups, ensuring that the classifier makes errors at comparable rates for all groups. This objective is appropriate in the medical context because both types of errors can have important consequences. False negatives may delay diagnosis and treatment, while false positives can lead to unnecessary follow-up procedures, anxiety, or additional healthcare costs.

The results show that ThresholdOptimizer substantially changes the behavior of the classifier compared to the tuned Random Forest. Recall increases dramatically when using ThresholdOptimizer. While the tuned Random Forest predicts almost no positive

instances (recall near zero), the ThresholdOptimizer achieves recall values around 0.10–0.15 across splits. This indicates that the post-processing algorithm successfully adjusts decision thresholds to identify more positive cases. However, this improvement comes with tradeoffs: Accuracy decreases from approximately 0.88–0.89 (tuned RF) to around 0.83–0.85. False positive rate (FPR) also increases noticeably.

In terms of fairness metrics, the ThresholdOptimizer produces larger but reasonable disparities than the tuned Random Forest for several measures, including FNR difference, FPR difference, and selection rate difference. The tuned Random Forest already exhibited near-zero fairness gaps, but it is primarily because it predicted almost all instances as negative.



The gender-specific plots show that the ThresholdOptimizer increases recall and false positive rates by similar magnitudes for both female and male groups. It is consistent with the equalized odds constraint, which aims to align both FNR and FPR across demographic groups.

(e)

Several general patterns emerge from the comparison of the different fairness-enhancing interventions.

First, fairness metrics can appear favorable even when predictive performance is poor. Both the tuned Random Forest and the Adversarial Fairness Classifier achieved

near-zero disparity metrics primarily because they predicted the negative class for most instances. This demonstrates that small fairness gaps do not necessarily imply meaningful or useful model behavior, especially under strong class imbalance.

Second, fairness interventions do not influence model behavior equally. The adversarial method showed limited impact on performance across different values of the fairness parameter. In contrast, the post-processing ThresholdOptimizer produced substantial changes in recall and error rates, indicating that threshold adjustment may be more effective when the base classifier is already trained and when the dataset is imbalanced.

Third, improving fairness or utility often requires sacrificing accuracy. The ThresholdOptimizer increased recall and reduced false negative rates but lowered overall accuracy and increased false positives. This highlights a fundamental trade-off between identifying more positive cases and maintaining high overall correctness.

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